

Presentation
for
6th Semester Optional Project (EC 300)
on
Swarm Environment Monitoring
Using Firebird VI Robots
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Presentation Outline

- ✓ **Introduction:** Mobile Robotics & FireBird VI
- ✓ **Literature Review:** APF & Swarm Optimization
- ✓ **Motivation:** Hardware Drift & Model Mismatch
- ✓ **Methodology:** Centralized Architecture
- ✓ **State Machine:** Autonomous Workflow Logic
- ✓ **Results:** Simulation vs. Hardware
- ✓ **Conclusion:** Summary of Findings
- ✓ **Future Work:** Concurrent Navigation

Autonomous Swarm Robotics

Mobile Robotics integrates sensing, computation, and actuation to perform tasks in dynamic surroundings.

- ✓ Shift from single-agent to **collective behavior** systems.
- ✓ **FireBird VI Platform:** Research-grade robots with Intel NUC processing.
- ✓ ROS1 Noetic framework for communication.
- ✓ **Significance:** Enables 24/7 warehouse operations and hazardous zone exploration.



Seminal Research & Context



Target Tracking

Modified Artificial Potential Fields (APF) offer efficient navigation but struggle with local minima and localization drift.

— Sarma & Panja (2025)



Optimization

Methods like ADMM coordinate large-scale systems but face hardware constraints like communication delays.

— Shorinwa et al. (2024)



Grid Mapping

Gmapping (FastSLAM) provides robust static maps, while Adaptive Monte-Carlo Localization (AMCL) handles dynamic pose estimation.

— Grisetti et al. (2007)

Motivation & Problem Statement

Objective: Develop a robust, centralized Master-Slave network architecture in ROS1 Noetic that automatically compensates for kinematic errors and synchronizes coordinate frames.

19.5%

Systematic Rotation Error

Theoretical optimization often fails due to **real-world hardware discrepancies**.

- ✓ **Localization Drift:** Position errors caused by wheel slip.
- ✓ **Local Minima:** Potential field navigation failures in unknown environments.
- ✓ **Model Mismatch:** TurtleBot3 models used previously have incorrect wheel track widths.

Methodology

System Architecture & Control Logic

Centralized System Architecture



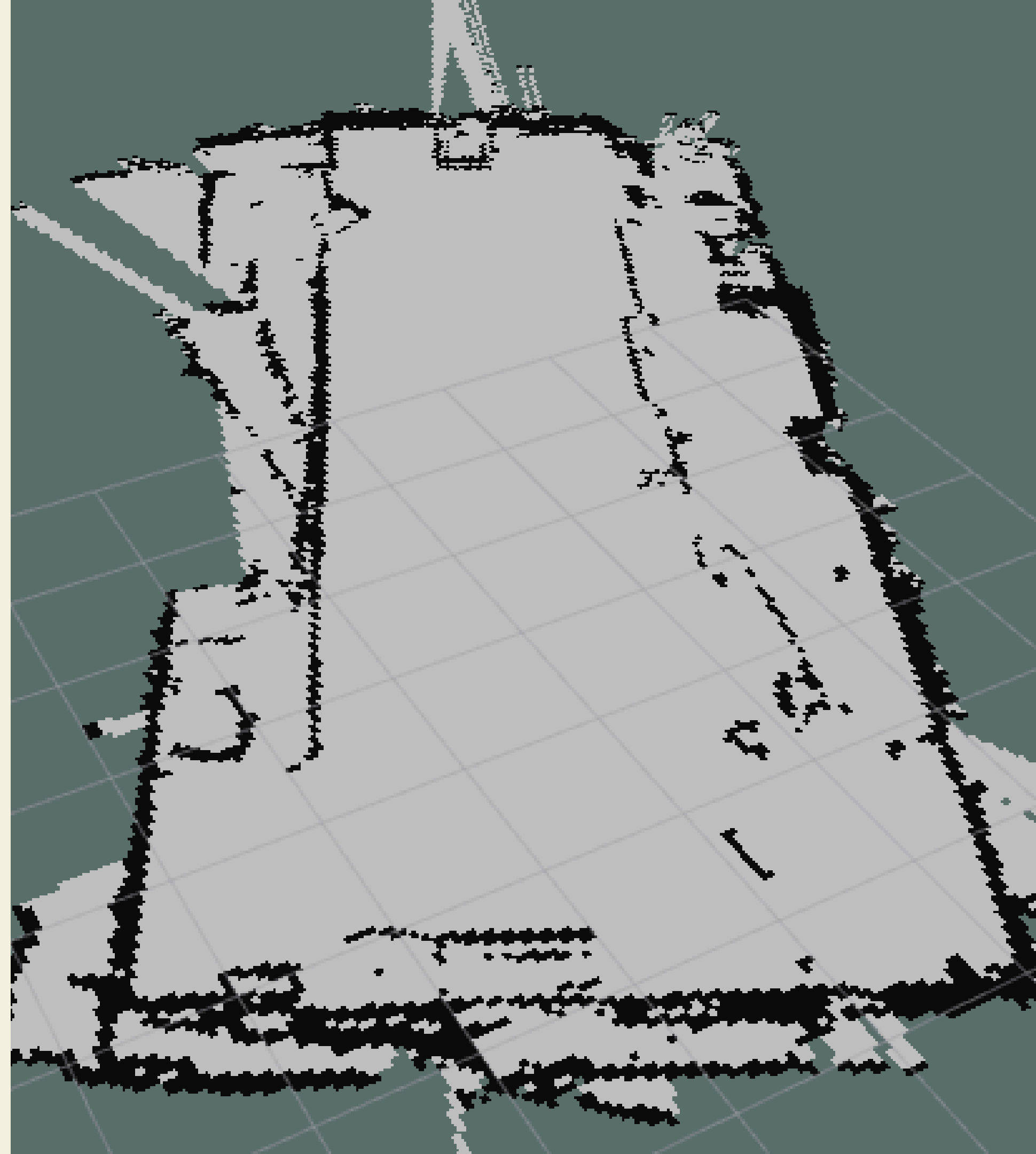
Hierarchical Control Layers

- ✓ **Command Layer (Remote PC)**: Global mapping, goal dispatching, and Master ROS core.
- ✓ **Execution Layer (Firebird NUC)**: Low-level actuation and sonar-based safety checks.
- ✓ **Namespace Isolation**: Independent topics for /robot1, /robot2, etc.

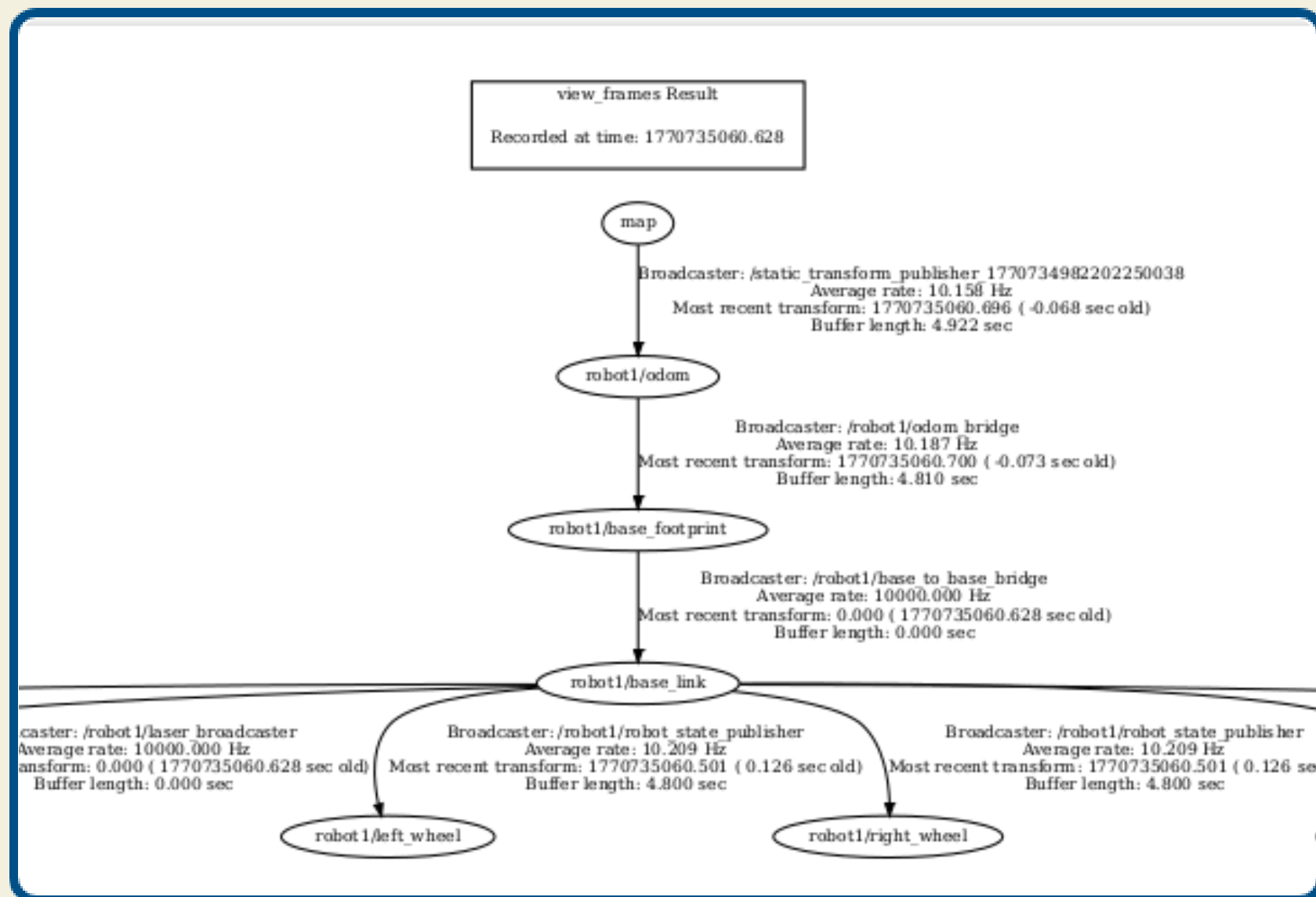
Environment Mapping

Arena mapping using **Gmapping (FastSLAM)** algorithm.

Occupancy grid generation enables accurate path planning and obstacle detection.



TF Tree Diagnostics



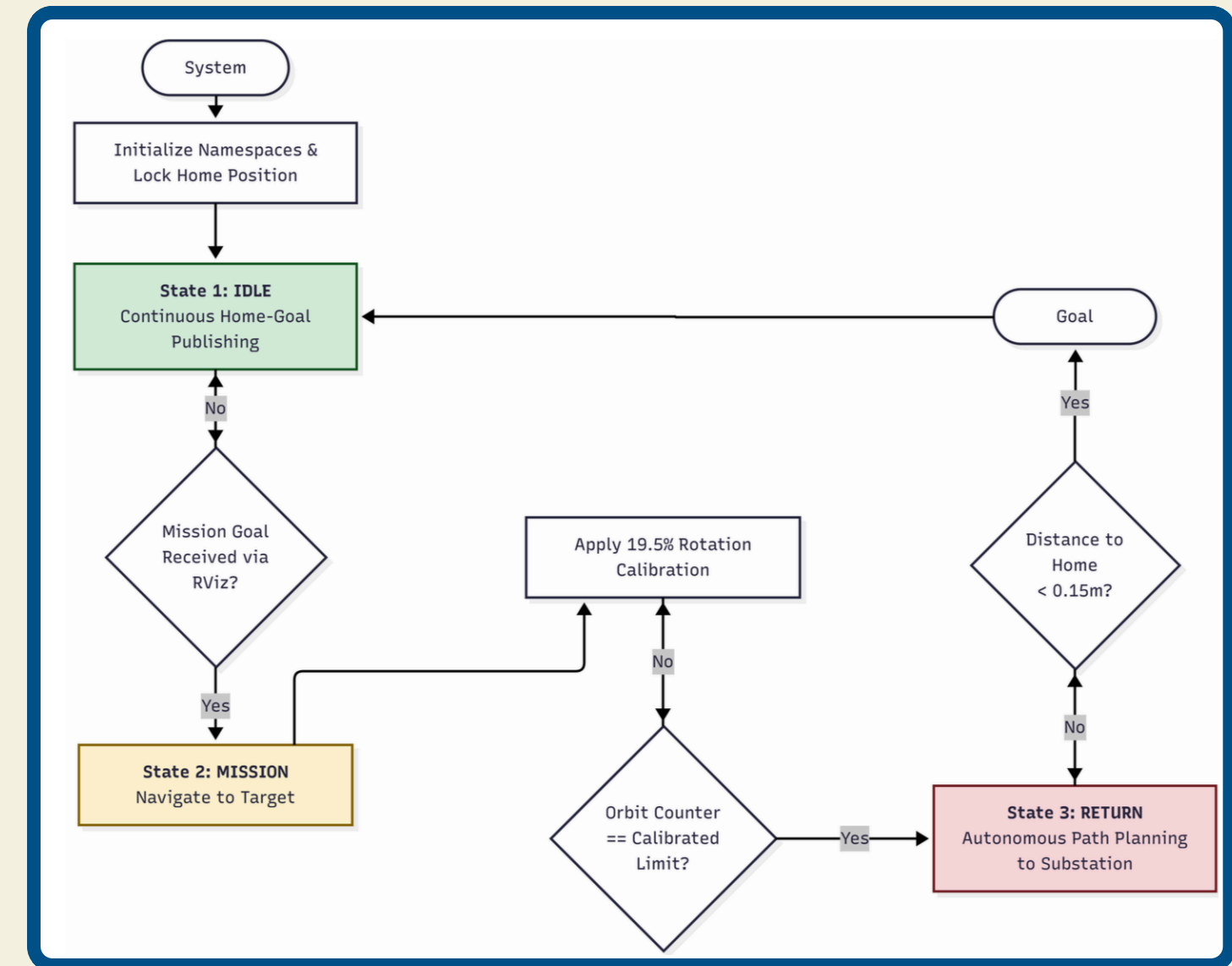
Unified coordinate frames are essential for swarm collision avoidance.

- ✓ **Custom Universal AMCL:** Manual broadcast of map-to-odom transforms.
- ✓ Global **Map** frame as root parent.
- ✓ Ensures all agents share a consistent world view.

Mission Logic State Machine

The **CalibratedScoutCommander** manages mission lifecycle through sequential state transitions:

- ✓ **IDLE:** Locks home coordinates to prevent odometry wandering.
- ✓ **MISSION:** Executes circular patrolling based on orbit counters.
- ✓ **RETURN:** Autonomous path planning back to station threshold (0.15m).



Hardware Compensation Model

A systematic over-rotation of 19.5% was measured. We integrated a calibration factor into the navigation stack to adjust the angular limit:

$$\theta_{\text{cal}} = \frac{2\pi}{1 + \text{error_ratio}} \approx 301.26^\circ$$

This adjustment ensures spatial consistency when robots transition between shared occupancy grids.

Mission State Machine

STATE 1

IDLE

Continuous home-goal publishing to lock coordinates at the substation.



STATE 2

MISSION

Nearest agent dispatched.
Navigate to target and execute calibrated 360° scouting maneuver.

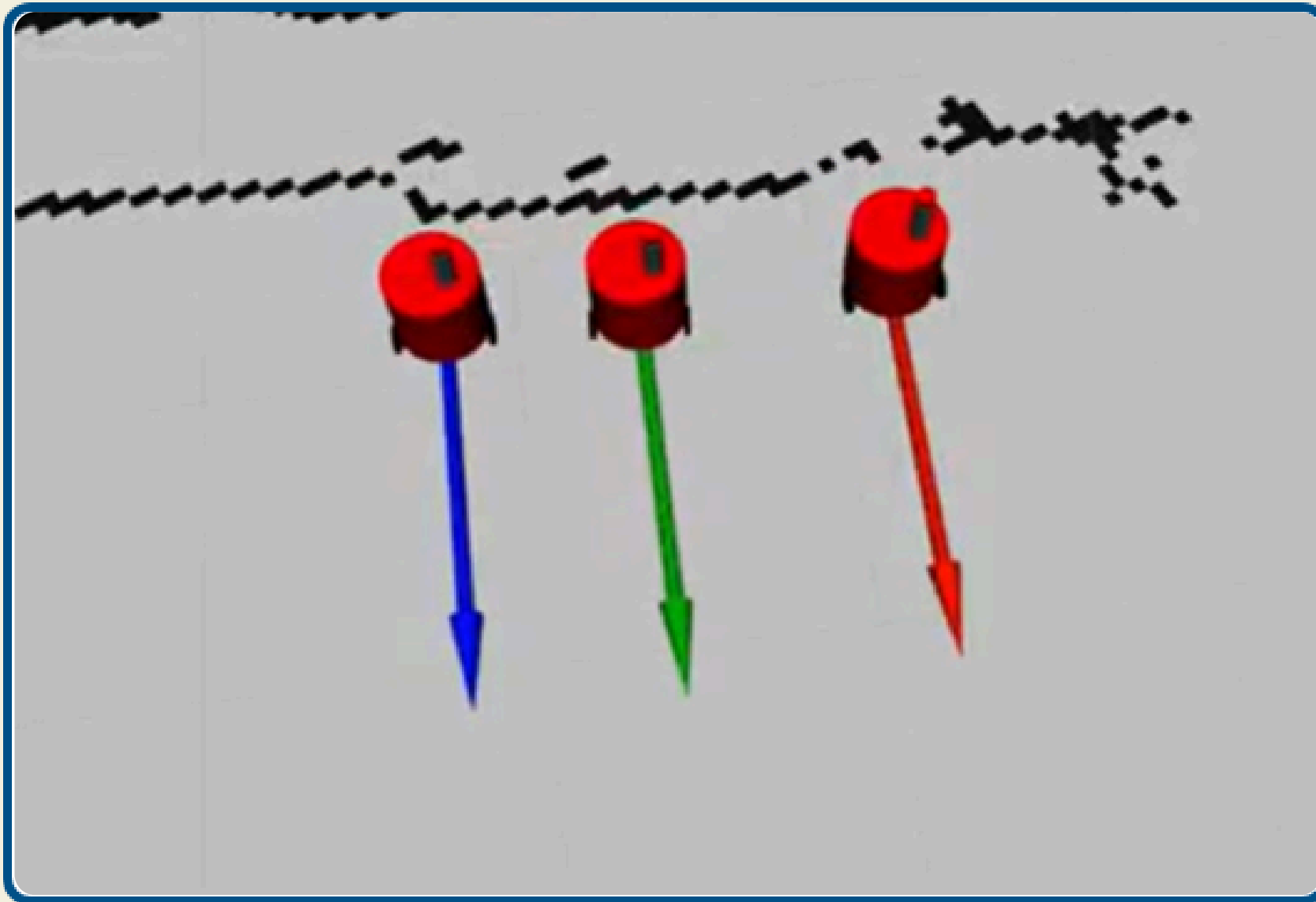


STATE 3

RETURN

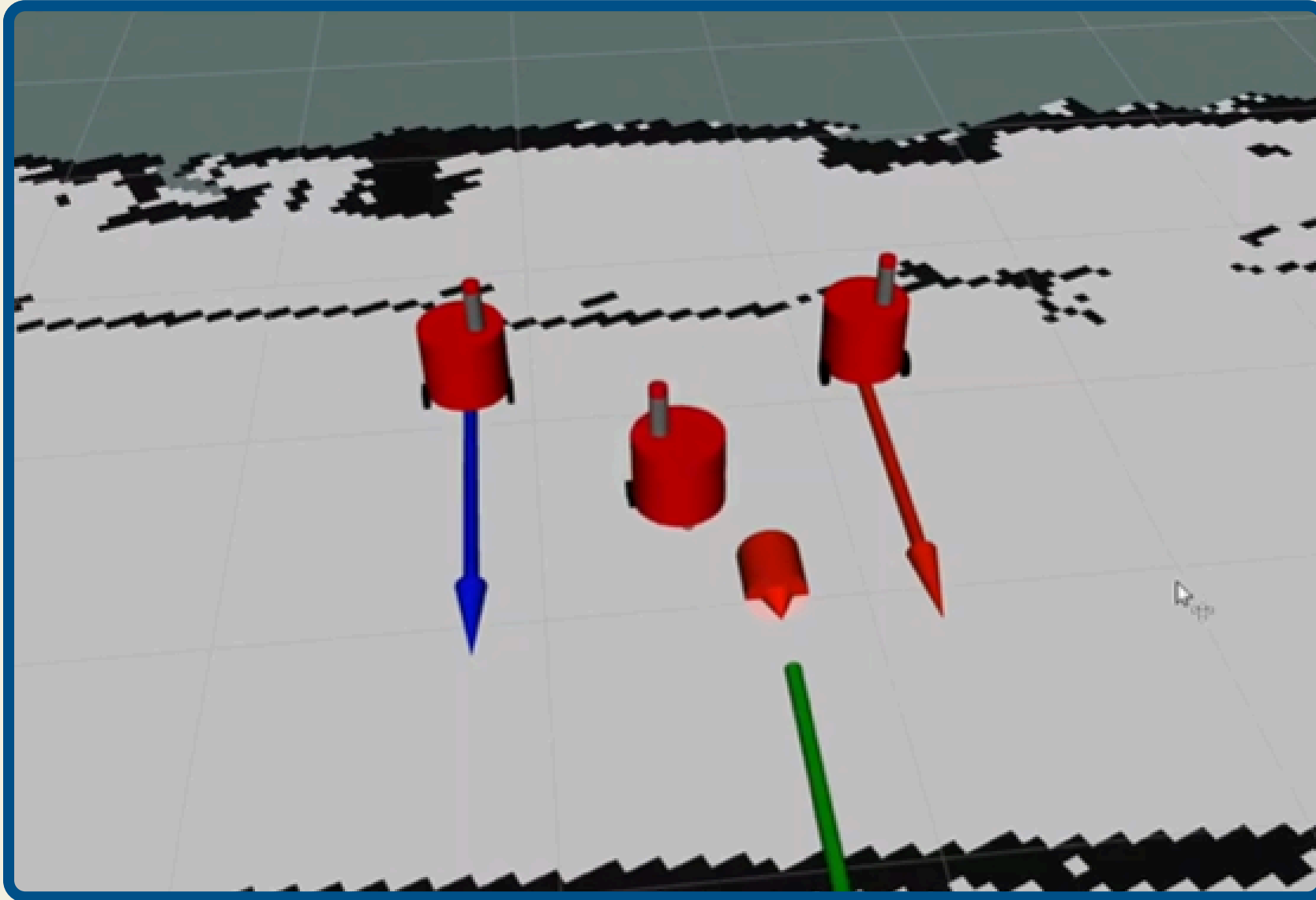
Autonomous path re-planning to substation. State resets to IDLE only when within a 0.15m arrival threshold.

State 1: IDLE Stability



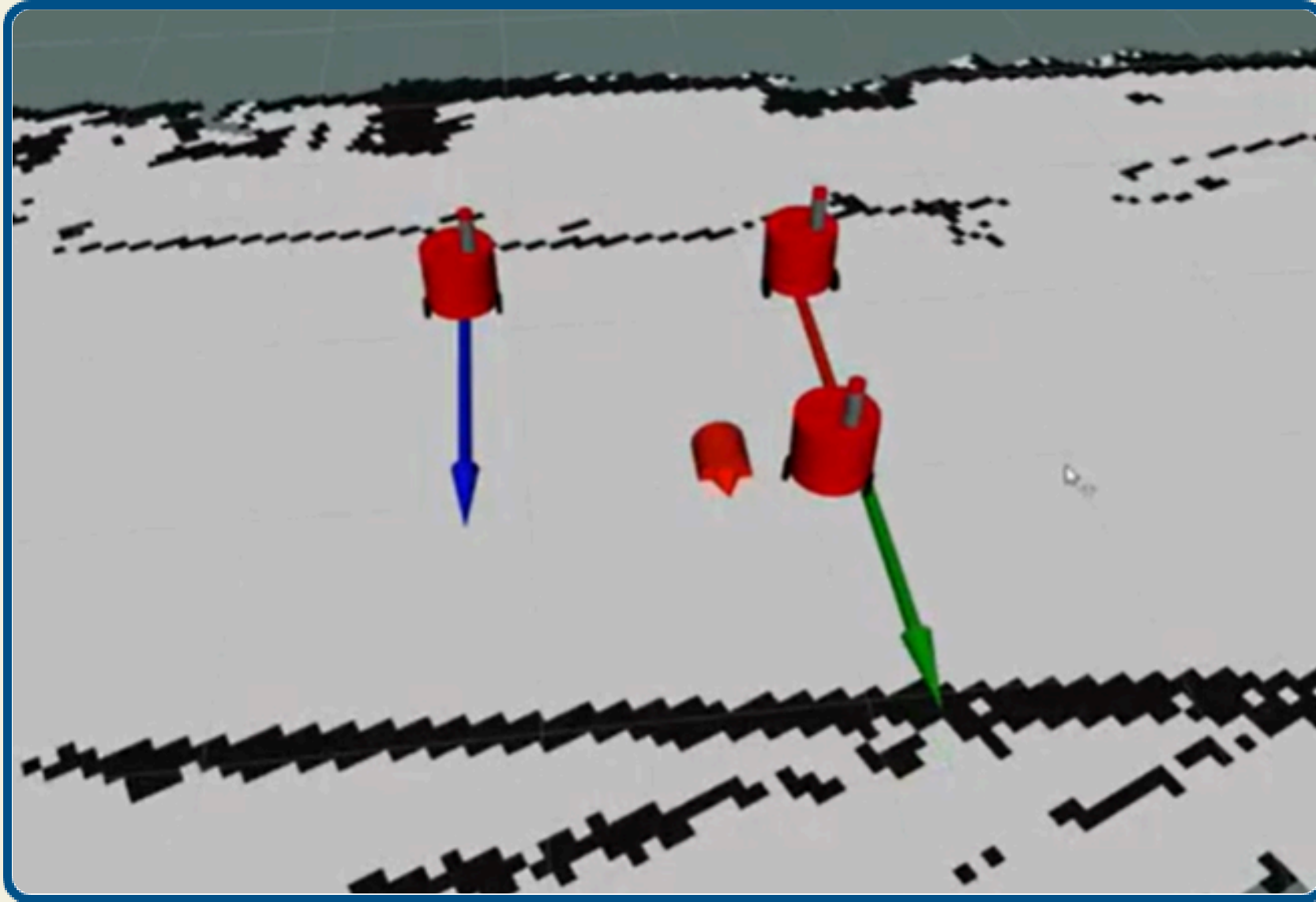
Performance: Positional variance maintained $< 0.05\text{m}$ at substation hubs.

State 2: *MISSION* Execution



Orbit Radius: Stable 0.5m scout maneuvers achieved after rotation calibration.

State 3: RETURN Performance

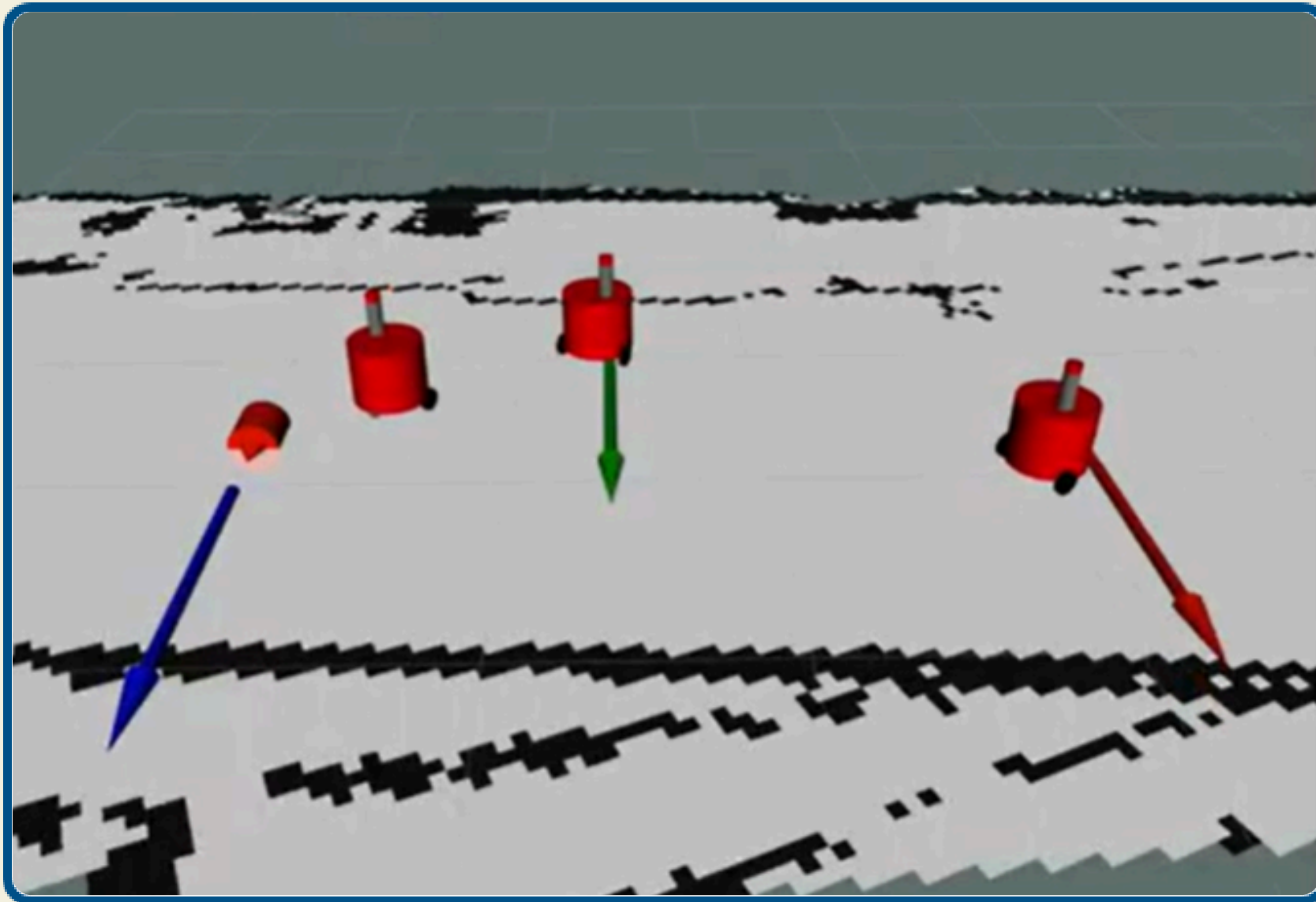


Recovery: Robots successfully returned to locked home coordinates within 0.15m precision.

DEMO

System Demonstration

Dual MISSION Performance



Recovery: 2 Robots successfully arrived and patrolled near mission coordinates within 0.15m precision.

Conclusion

Summary

- ✓ Established robust centralized ROS Master-Slave architecture.
- ✓ Developed mathematical rotation calibration for Firebird VI hardware.
- ✓ Validated reliable RTB (Return-to-Base) mission protocols.
- ✓ Custom AMCL Node and mathematical compensation model effectively neutralized the 19.5% hardware drift.

Future Research Directions

Future Work

- ✓ **Prioritized Planning:** Developing a centralized traffic manager to assign priority ranks to agents, thereby resolving deadlocks in narrow environments
- ✓ **Consensus-Based Formations:** Implementing Laplacian-based swarm dynamics to allow the formation to shift dynamically in response to obstacles.

Key References

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Thank You

Any Questions?

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